

SECR3253 NETWORK PROGRAMMING GROUP ASSIGNMENT (5%)

(1) Objective:

Students are required to work in groups to complete a small network programming or automation project. The emphasis of this assignment is on teamwork, communication, collaboration, and contribution by all group members.

(2) Group Size: 5 students per group (can mix across sections)

(3) Project Scope:

(a) Develop an automation solution using Docker, Ansible, NETCONF or a combination of these technologies. The project must be developed collaboratively using Git and/or GitHub.

(b) The automation solution must perform the following device configuration tasks:
Configure IP address, user account, banner message, interface description, static route and retrieve device information

(c) Automatically collect and display the following Linux system information:
hostname, current date and time, CPU information, memory usage, disk usage, logged-in users, top 5 running processes by CPU usage.

(4) Collaboration Requirements:

- The project must be managed using a GitHub/Git repository.
- All group members must contribute through commits.
- Commit history should demonstrate active participation from all members.

(5) Deliverables:

(a) GitHub/Git Repository URL

(b) 2 pages of Personal Reflection Report:

- Describe your contribution to the project
- Discuss challenges encountered during collaboration
- Describe what you learned from the project

(6) Dateline: **9 AM. 6 JULY 2026**

(7) Marking Criteria: Project Completion and Functionality 30%, Git Repository and Documentation 20%, Collaboration Evidence (Git Commits) 30%, Personal Reflection Report (20%)

(8) Academic Integrity:

All submitted work must be the group's own work. Copying from other groups is prohibited. AI tools may be used for learning, debugging, and code explanation. Students must understand and be able to explain all submitted work.